# Ângelo Azevedo

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Computer Science student specializing in **cybersecurity**, focusing on **security engineering** and **vulnerability research**. **Scout since age 7**, drummer, and skimboarder, with an active interest in **cinematography** and **multimedia**. Practical background in **forensics**, **incident response**, and **platform hardening** across red-team audits, academic labs, and CTFs.

## Education

**Instituto Superior Técnico** *2025 - Present*
M.Sc. in **Computer Science and Engineering** *Lisbon, Portugal*
*Specialization in **Cyber-Security & IT Systems*** **GPA 17/20**

**Instituto Superior de Engenharia de Lisboa** *2022 - 2025*
B.Sc. in **Computer Science and Computer Engineering** *Lisbon, Portugal*

## Experience

**Teaching Assistant (CSF)** *Aug 2026 - Nov 2026*
*Instituto Superior Técnico (IST)*
- Teaching Assistant for the **Cybersecurity Forensics (CSF)** course in the Técnico Computer Science M.Sc. program, designing lab exercises and grading final projects for 100+ students.

**Forensics Instructor** *2026*
*Security Team @ Técnico (STT)*
- Taught hands-on **memory and disk forensics** workshops for 30+ STT members using Volatility and Autopsy (*link*).

**Red Team & Security Auditor** *Feb 2026 - Present*
*Er3bUs Security Team (Técnico)*
- Led a 5-member team conducting a penetration test and code audit on GameCourse, reporting 4 critical staging vulnerabilities and remediation patches.

**Course Instructor & Research Grantee** *Oct 2025 - Present*
*Instituto Superior Técnico (IST)*
- Instructor for the **Programming & Electronics** course (OASIS) and **Oeiras Educa** workshops, teaching **cryptography**, **steganography**, **metadata analysis**, **OSINT** and **hash cracking** alongside **Arduino** and **IoT**.

**Programming Teacher** *Dec 2024 - Jun 2025*
*SharkCoders*
- Taught **Python**, algorithmic thinking, and robotics to classes of 15+ students aged 8 to 17.

## Projects

**root-my-nothing (Android Vulnerability Tooling)** *2026*
- Ported the **CVE-2026-43499 (GhostLock)** futex vulnerability to Nothing Phone (1), extracting kernel struct offsets with pahole and building an Android NDK and Kotlin app to deploy KernelSU.

**FACT (Forensic Artifact Creation Toolkit)** *2026*
- Engineered an automated simulation framework using **Python**, **Packer**, **QEMU** and **PowerShell** to generate multi OS forensic labs, forging realistic attack timelines (**Registry**, **BitLocker**, **ActivitiesCache**, memory dumps) without evidence contamination for **100+ students**.

**Voltage OS Android & Kernel Trees** *2025 - Present*
- Maintained **Linux 5.4 kernel** and device trees for Nothing Phone (1), configuring custom builds, **KSU-SUSFS** root-hiding integration, and hardware abstraction (HAL) for an active community of **20+ users**.

**Samsung Galaxy Book Linux Hardware Fixes** *2026*
- Contributed to Linux hardware support on Galaxy Book laptops by automating **Secure Boot module signing** via Kbuild, developing **systemd runtime services**, and building userspace camera controls.

**Console Binary Reverse Engineering (Switch)** *2026*
- Reverse-engineered ARM64 binaries (**Mario Kart 8 Deluxe**) in **IDA Pro**, isolating NVN capability checks and using **pwndbg** over an **Atmosphere GDB stub** to hijack the UI state machine and test enabling Switch 2 features on Switch 1.

**NixOS Security Workstation & NUR Packages** *2024 - Present*
- Engineered a **declarative NixOS security workstation** with modular environments for forensics and reverse engineering, maintaining **95+ security packages** in NUR (*link*) including custom Volatility 3 plugins and LEAPP triage tools.

## Awards

**Academic Excellence Award**
*2022*
Received excellence award for **best new students**

## Skills & Languages

**Security & Forensics** Wireshark, Burp Suite, Ghidra, IDA, Volatility, Autopsy, YARA, Frida, GDB

**Programming & Scripting** Python, Bash, C, Go, Java, Kotlin, JS/TS, SQL, Nix

**Systems & Infrastructure** Linux, Android (AOSP), NixOS, Windows, Docker, Git, CI/CD, Ansible

**Languages** **Portuguese** (Native), **English** (Fluent)

## Cybersecurity Competitions

**ECSC & Cybersecurity Challenge PT**
*2026*
- Selected as **Team Captain** for the **National Team** at the 2026 ECSC (*link*), having qualified through the CSCPT (*link*) to represent **Portugal**.

**BSides Lisbon CTF** *2025*
- **2nd Place** - Solved forensics, web, and cryptography challenges in a 24-hour competition on the biggest portuguese cybersecurity event.

**Security Team @ Técnico (STT)** *2025 - Present*
- Técnico's Student CTF Team (#1 in Portugal, 245th worldwide). Active international competitor.
- Focus on **forensics**, **web**, and **reverse engineering** challenges.

**r4ph3cks CTF Team** *2022 - Present*
- **Founder & Leader** of university CTF team.

**Personal CTF Write-ups** *Present*
- Publishing detailed **CTF write-ups** and **scripts** at *github.com/ang3lo-azevedo/ctfs*.

**Self-Hosted Infrastructure & Networking** *2024 - Present*
- Administering a **Debian-based homelab** running **Docker/Portainer** to deploy and manage self-hosted privacy services (Nextcloud, Immich, SearXNG).
- Configured an **NGINX reverse proxy** for secure external access and implemented an automated **NAS backup strategy**.